

PEAS Framework and Task Environment Analysis

System: Agricultural Monitoring Drone for Crop Disease Detection

TASK 1: PEAS SPECIFICATION

Performance Measures:

- Disease detection accuracy
- False positive/negative rate
- Field coverage efficiency
- Early detection time
- Battery efficiency
- Data transmission reliability

Environment:

Agricultural fields, crops, soil conditions, weather (wind, rain, humidity), pests, diseases, obstacles, farmers, and infrastructure.

Actuators:

Flight motors, camera gimbal, GPS navigation, wireless communication system, alert system.

Sensors:

RGB camera, multispectral sensor, thermal camera, GPS, IMU, LiDAR/ultrasonic sensors, weather sensors.

TASK 2: ENVIRONMENT CLASSIFICATION

Observability: Partially Observable - Some crop conditions are hidden or not directly visible.

Agents: Single-Agent - One drone performs monitoring.

Determinism: Stochastic - Weather and disease spread introduce uncertainty.

Episodic/Sequential: Sequential - Each action affects future decisions.

Static/Dynamic: Dynamic - Environment changes continuously.

Discrete/Continuous: Continuous - Movement and sensing are continuous.

TASK 3: CRITICAL ANALYSIS

1. Most Challenging Dimension:

Partial observability is the biggest challenge because crop diseases may be hidden or unclear. The drone must infer from

2. Utility Function:

$$U = 0.5 \cdot \text{Accuracy} + 0.3 \cdot \text{Coverage} - 0.2 \cdot \text{Energy}$$

If energy weight increases, the drone prioritizes battery saving, shorter routes, and reduced scanning frequency.

TASK 4: JSON STRUCTURE

```
{
  "system_name": "Agricultural Monitoring Drone",
  "peas": {
    "performance_measures": ["accuracy", "false_rate", "coverage", "energy_efficiency"],
    "environment": "farmland with weather variability",
    "actuators": ["motors", "gps", "camera_control"],
    "sensors": ["rgb", "multispectral", "thermal", "gps", "lidar"]
  },
  "environment_classification": {
    "observability": "partially observable",
    "agents": "single-agent",
    "determinism": "stochastic",
    "sequential": "sequential",
    "dynamic": "dynamic",
    "continuous": "continuous"
  },
  "utility_function": "U = 0.5A + 0.3C - 0.2E"
}
```